

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10

Student Assessment Document

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10: THE PERIODIC TABLE (SUB-STRAND 2.1)

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Final Explanation: Why do fertiliser bags carry such specific formulae? Think about the two bags a Grade 10 learner in Kericho saw on her family's tea farm: CAN, $\text{Ca}(\text{NO}_3)_2 \cdot \text{NH}_4\text{NO}_3$, and DAP, $(\text{NH}_4)_2\text{HPO}_4$. Both are white crystalline solids that dissolve in water, and both contain nitrogen, yet their formulae are different. Answer each prompt in the blank space under it, using what you learned in Lessons 1 to 8. Write in full sentences. Where a question asks you to show working, write each step. You may draw dot-and-cross diagrams or small tables. Use the words you met in class (valence electrons, group, period, ion, valency, radical, charge balance) and link each answer back to the fertiliser bags.

Part 1: How the periodic table is organised

A learner in Kericho says: 'The periodic table is just a list of elements. It has nothing to do with the fertiliser on our tea farm.' Use calcium (found in CAN) and nitrogen (found in CAN and DAP) to show that she is wrong. In your answer: (a) explain what the periodic law is and how you discovered it when you sorted the 20 element cards; (b) explain what the periods and the groups of the periodic table tell you about an atom's electron arrangement; (c) state which group calcium and nitrogen are in (use the group numbers 1 to 8 for the main groups, as in Lessons 4 and 5) and give the reason for each.

Write your answer here:

Part 2: Why electron arrangement is significant

Explain why the arrangement of electrons in an atom, especially the outer shell, decides what ion the atom forms. Use calcium (2,8,8,2) and nitrogen (2,5) as your two examples. In your answer: (a) say what a full outer shell means for the stability of an atom and name one noble gas example; (b) explain, with the electron changes, how calcium forms its ion and how nitrogen forms its ion, giving each ion's symbol and charge; (c) explain why calcium forms a 2+ ion and not a 1+ or 3+ ion, using the noble gas rule.

Write your answer here:

Part 3: From ion charge to the formula $\text{Ca}(\text{NO}_3)_2$

Use the idea of valency and the criss-cross rule to write the formula of calcium nitrate. In your answer: (a) state the charge of the calcium ion and of the nitrate radical, and say what valency means; (b) show every step of the criss-cross method, writing the subscript that each symbol receives; (c) say how you decided whether the NO_3 part of your formula needs brackets; (d) check your formula by showing the total positive charge and the total negative charge.

Write your answer here:

Part 4: Why CAN and DAP have different formulae

Both fertiliser bags contain nitrogen, but the formulae on the two bags are different. Look at the NH_4NO_3 part of the CAN formula and at the DAP formula, $(\text{NH}_4)_2\text{HPO}_4$. In your answer: (a) give the charge of NH_4^+ and of NO_3^- and decide what ratio of these two ions gives an electrically neutral compound; (b) count the NH_4^+ ions in $(\text{NH}_4)_2\text{HPO}_4$ and work out the total positive charge they carry; (c) explain in one or two sentences what the two formulae have in common and why they still look different.

Write your answer here:

Part 5: Answering the driving question

Driving question: Why do the fertiliser bags on Kenyan farms carry such specific and different chemical formulae, and what does the inner structure of an atom have to do with it? Write a clear explanation for the learner in Kericho. Your answer must: (a) link the inner structure of the atom to the formula on the bag, putting the steps of your reasoning in the correct order; (b) use calcium and the CAN formula as your worked example; (c) use the balanced equation $\text{Na} + \text{Cl}_2 \rightarrow \text{NaCl}$ or $\text{Mg} + \text{O}_2 \rightarrow \text{MgO}$ from Lesson 7 to show what balancing does and why only coefficients, not subscripts, can be changed; (d) say in one sentence what you still wonder about.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Linking atomic structure to the periodic table	Clearly explains the periodic law, periods and groups, and correctly links calcium and nitrogen to their groups through their valence electrons.	Explains most ideas about periods, groups and valence electrons, with one small gap or error in placing calcium or nitrogen.	Describes the periodic table only as a list, or does not connect groups to valence electrons.
Explaining ion formation	Correctly shows electron loss or gain for Ca and N, gives Ca^{2+} and N^{3-} , and explains the noble-gas stability reason, including why the charge is fixed.	Gives correct ions and electron changes but the reasoning about stability or fixed charge is incomplete.	States ions without explaining the electron changes, or confuses cations and anions.
Writing and checking formulae	Applies the criss-cross rule accurately, uses brackets correctly for radicals, and verifies charge balance with numbers for $\text{Ca}(\text{NO}_3)_2$, NH_4NO_3 and $(\text{NH}_4)_2\text{HPO}_4$.	Writes correct formulae with the criss-cross rule but gives limited checking of charge balance or explanation of brackets.	Writes formulae incorrectly or without the steps, and does not check that charges balance.
Connecting the whole explanation to the phenomenon	Builds a complete, ordered causal chain from electron arrangement to valency to formula on the CAN and DAP bags, uses a balanced equation correctly to show conservation of atoms and why only coefficients change, and refers throughout to the Kericho learner.	Gives a mostly correct chain with one link missing or weakly explained, and refers to the fertiliser bags.	Gives disconnected statements and does not link atomic structure to the formulae on the bags.

ATTRIBUTION AND LICENSING

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